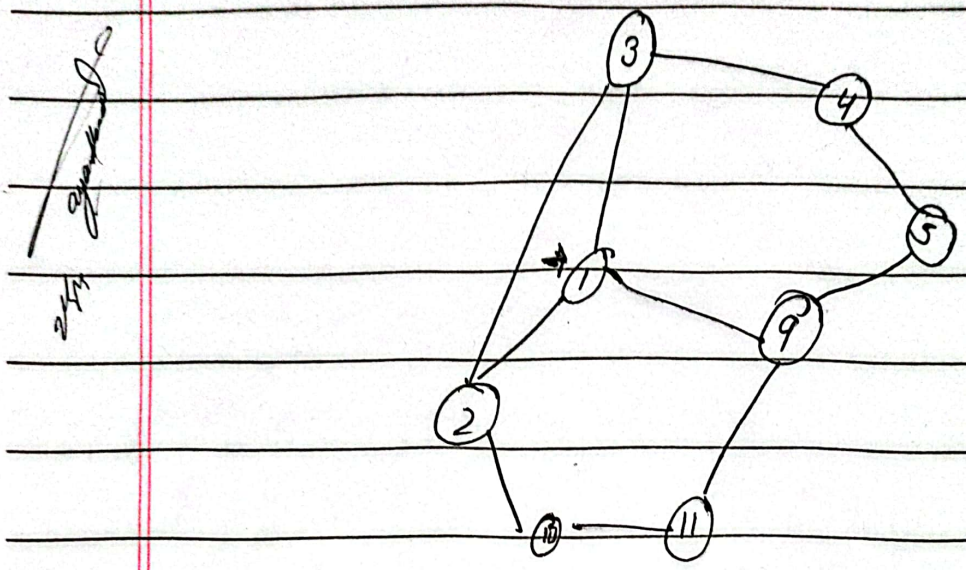
Graph traversal:

→ BFS (~~Depth~~ first search) => (level by level)



BFS:

→ we can start traversing from any point (until not mention in Q

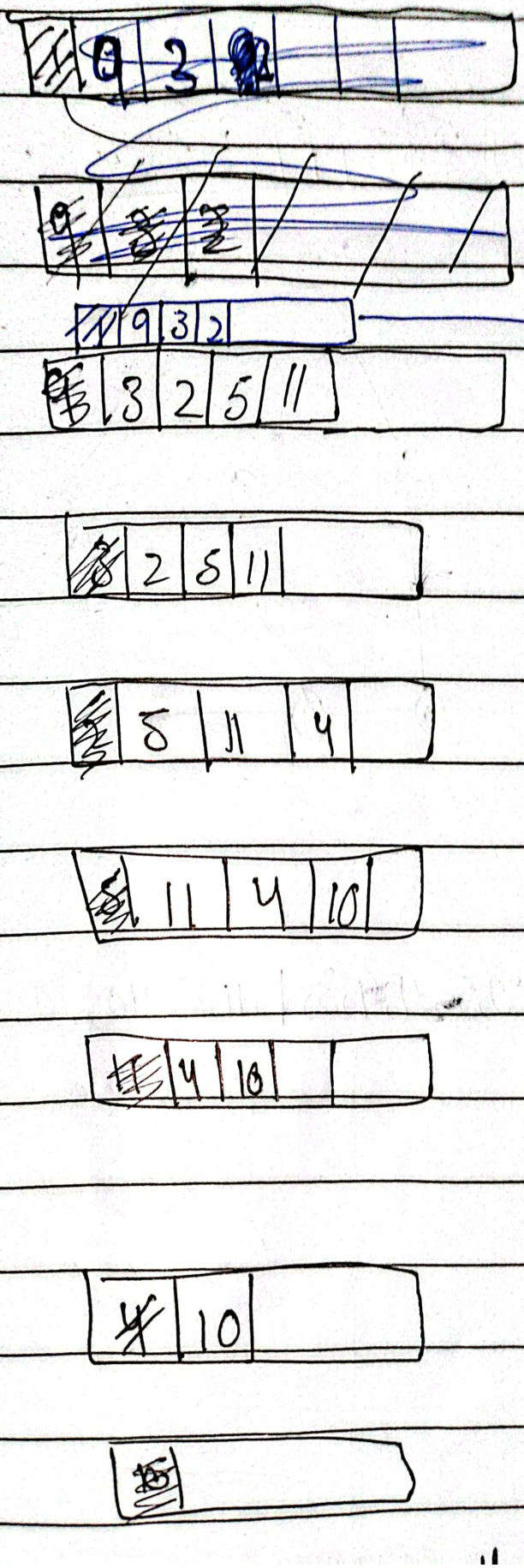
→ We can go anywhere.

→ We use que in BFS / visited que

Solution

Starting from ①

Date: _____



Output:

~~1, 9, 3~~

→ 1

: 1, 9

1, 9, 3

1, 9, 3, 2

1, 9, 3, 2, 5

1, 9, 3, 2, 5, 11

1, 9, 3, 2, 5, 11, 4

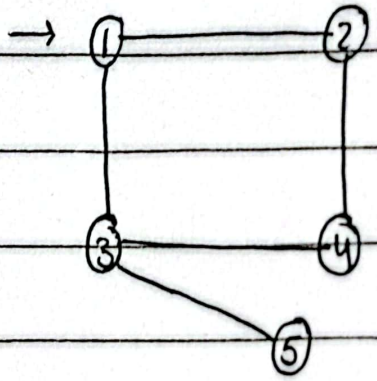
1, 9, 3, 2, 5, 1, 4

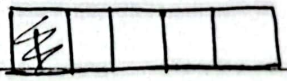
$\{1,2,3,4,5\}$

$\{(1,2),(1,3),(2,4),(3,4),(3,5)\}$

a) BFS

Start from 1:

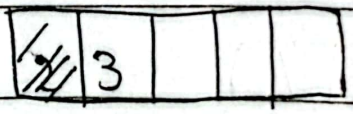




Output:

① is conected with

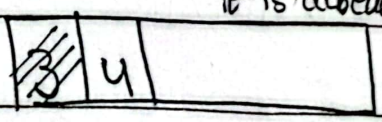
1



② is connected with

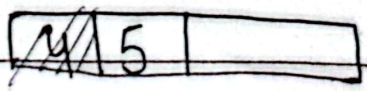
we will not write ① because it is already visited.

1,2



③ is conected with

1,2,3



Day: _____ Date: _____

Agarwal

we will not write 3, 1 because it is already visited

output:

1, 2, 3, 4,

As 5 is ~~connected~~ connected with 3 and 3 is already visited so we will not write is

1, 2, 3, 4, 5

Final output.

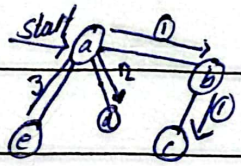
Day:_____ Date:_____

(DFS)

Depth - First search.

-> we can start traversing from any point. (until no mentioned)

-> we traverse all adjacent vertices one by one.



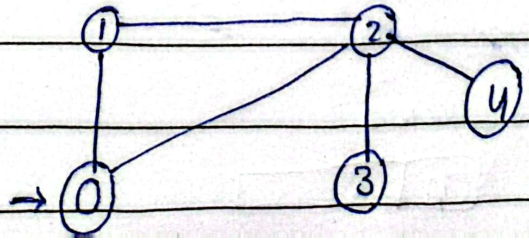
from given first way, we go from (a) towards (b) then (c) untill we reach ~~each each~~ end. ~~and~~ then came back to start point if there are more then one way.

second way is (a) to (d)

third way is (a) to (e).

-> we use stack for backward movement.

Example:



Day: _____ Date: _____

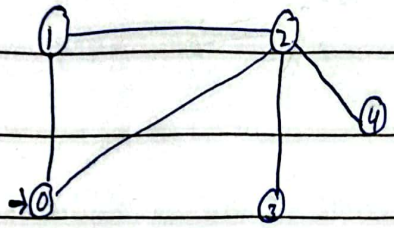
Output: 0 1 2 3 4

Explanation:

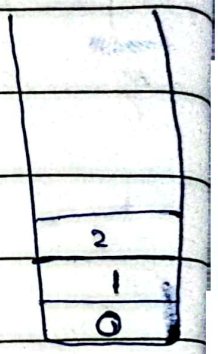
The source verten is 0

start at 0: we marked it as visited

output : [0]



stack



we have two ways

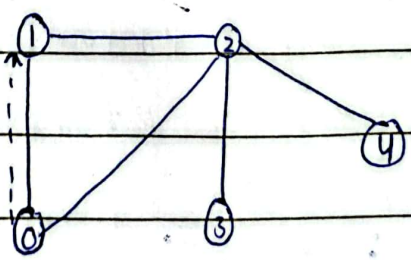
(0) to (1)

or

(0) to (2)

for all step we using this stack

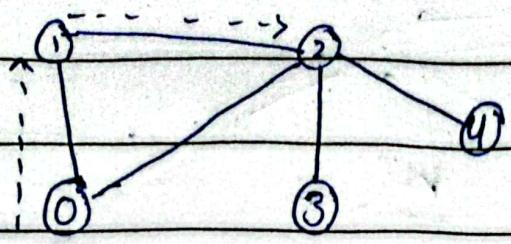
Move to 1:



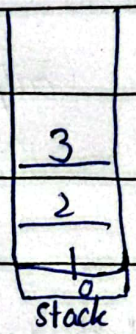
Mark 1 visited

output : [0 1]

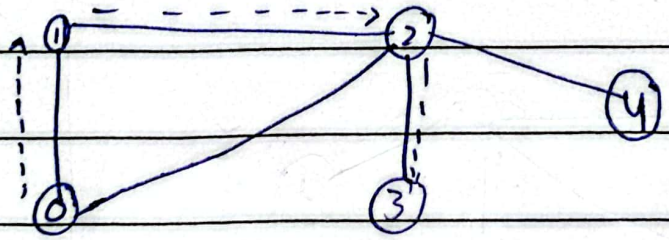
Move to 2:



output : 0 1 2



Move to 3:



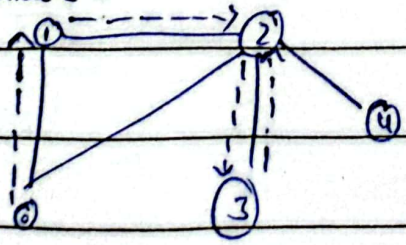
output : 0 1 2 3

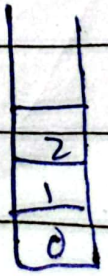
∴ now back track from 3 to 2

we use stack for this movement

because last in fist out we have the point where to go back so we remove 3 from stack.

Backtrack : 3

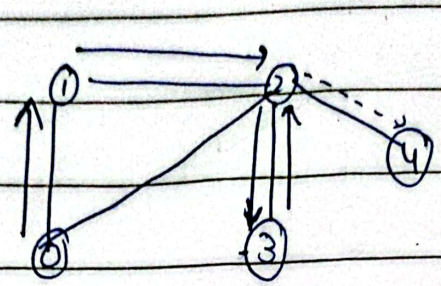


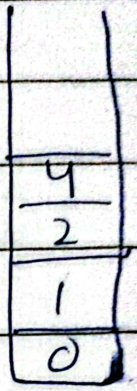


updated stack.

Day:

move to 4:

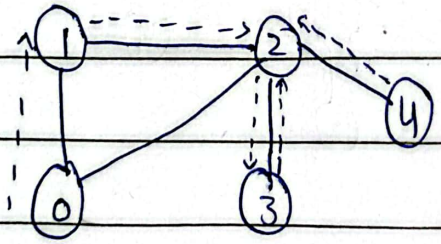


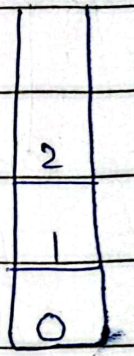


output : | 0 | 1 | 2 | 3 | 4 |

Agenttemas

back track from 4

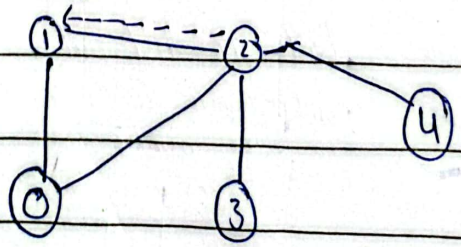




output will be same

| 0 | 1 | 2 | 3 | 4 |

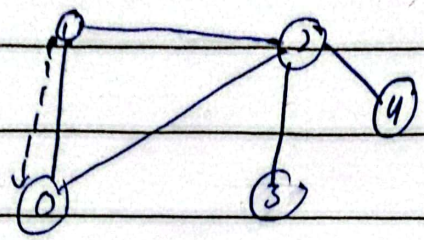
back tracking from 2

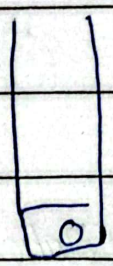


Day: _____ Date: _____

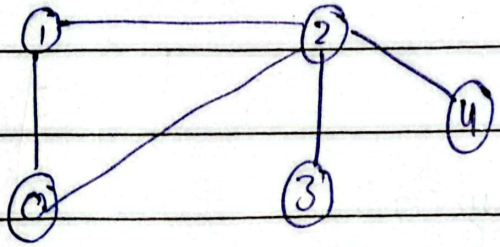
Ayan Ahmed

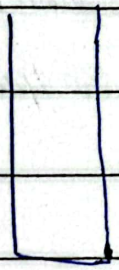
back tracking from 1





Back to source 0:





stack is empty.

Conclusion / justification of step:

start at 0: mark as visited, output: 0

Move to 1: " , output: 1

Move to 2 " , output: 2

Move to 3: " , output: 3 , backtrack to 2

Move to 4: " , output: 4 , backtrack to 2 then 1 then 0)

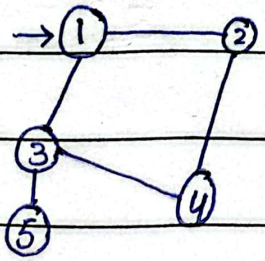
Day: _____ Ayantika Date: _____

Assignment Question:
(DFS)

vertices : { 1, 2, 3, 4, 5 }

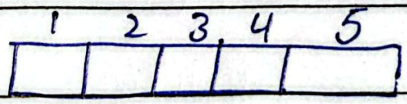
edges : { (1,2), (1,3), (2,4), (3,4), (3,5) }

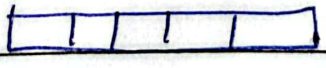
Graph:

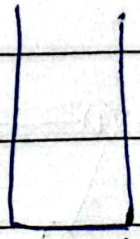


Here we have ending point 5 and 4, from where we are not going anywhere.

Solution

visited[] : 

output[] : 



stack (empty).

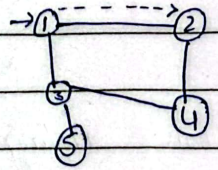
Day:

» Start from 1:

visited : | 1 | 2 | 3 | 4 | 5 |
| T | F | F | F | F |

output | 1 | | | | |

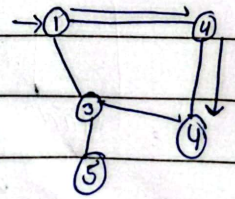
» move to 2



visited : | 1 | 2 | 3 | 4 | 5 |
| T | T | F | F | F |

output : | 1 | 2 | | | |

» move to 4:

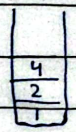


Ayat Ahmed

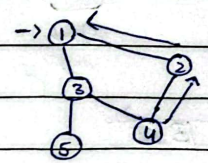
Date:

visited [ ]: | 1 | 2 | 3 | 4 | 5 |
| T | T | F | T | F |

output : | 1 | 2 | 4 | | |



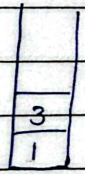
» backtrack from 4 to 1



» move to 3 :

visited : | 1 | 2 | 3 | 4 | 5 |
| T | T | T | T | F |

output: | 1 | 2 | 4 | 3 | |

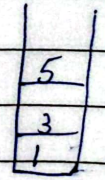


» now we will not move to 4 because it is already visited.

move to 5

visited | 1 | 2 | 3 | 4 | 5 |
| T | T | T | T | T |

output : | 1 | 2 | 4 | 3 | 5 |

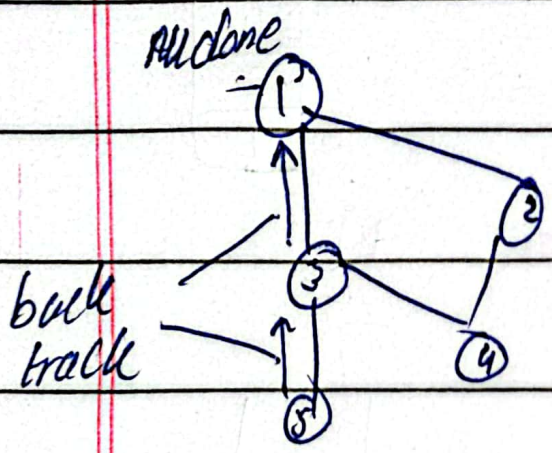


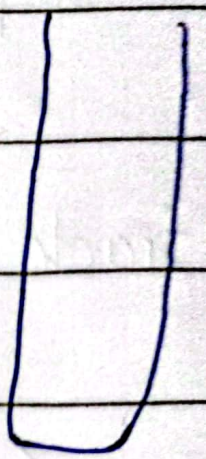
Day:_____

Ayesha Ahmed

Ayesha

•) back tracking from 5 to 1





stack empty

output : | 1 | 2 | 4 | 3 | 5 |

Ans